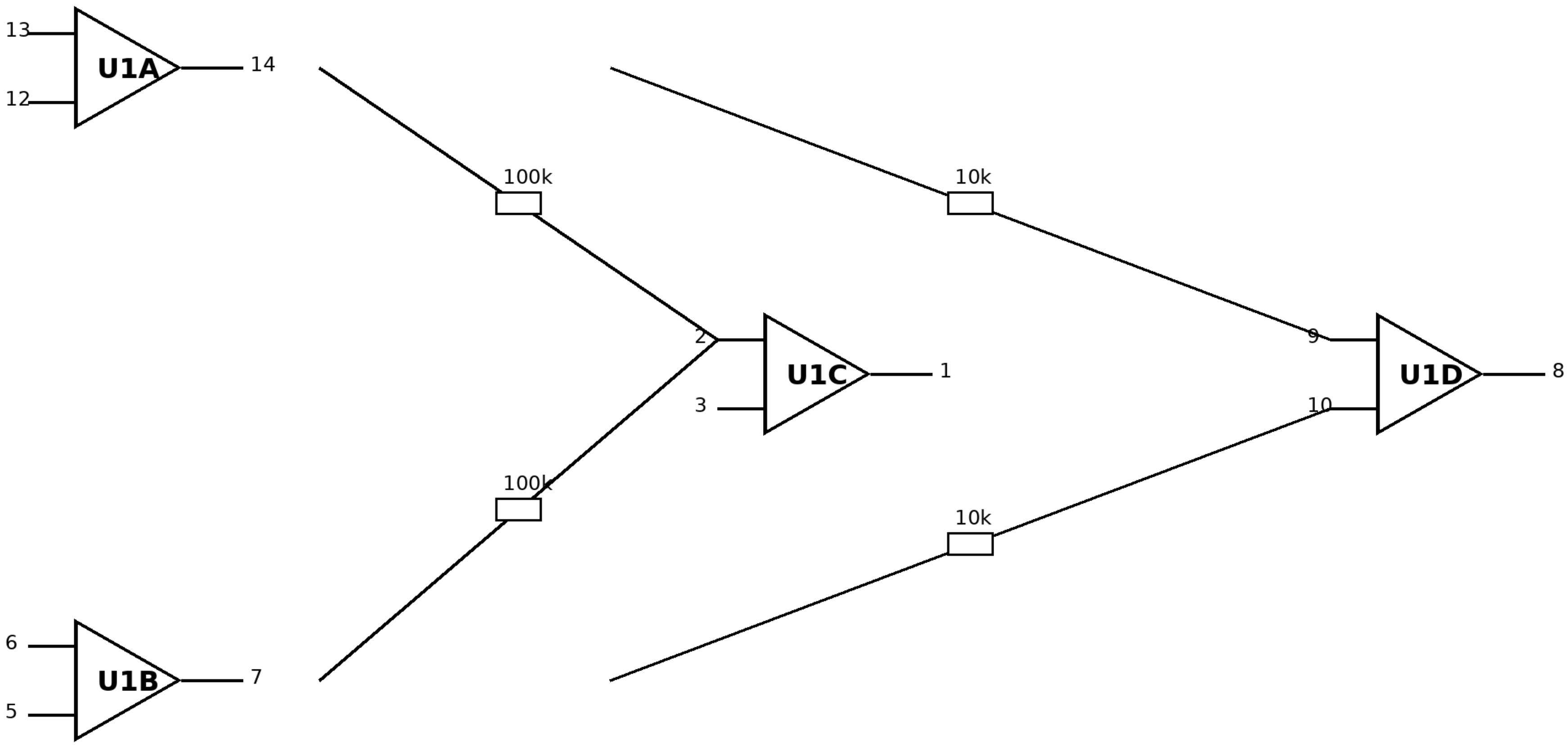
DIY EEG Guide (TL074 + 9V)

Professional EEG Schematic (10,000x Gain + Active DRL)



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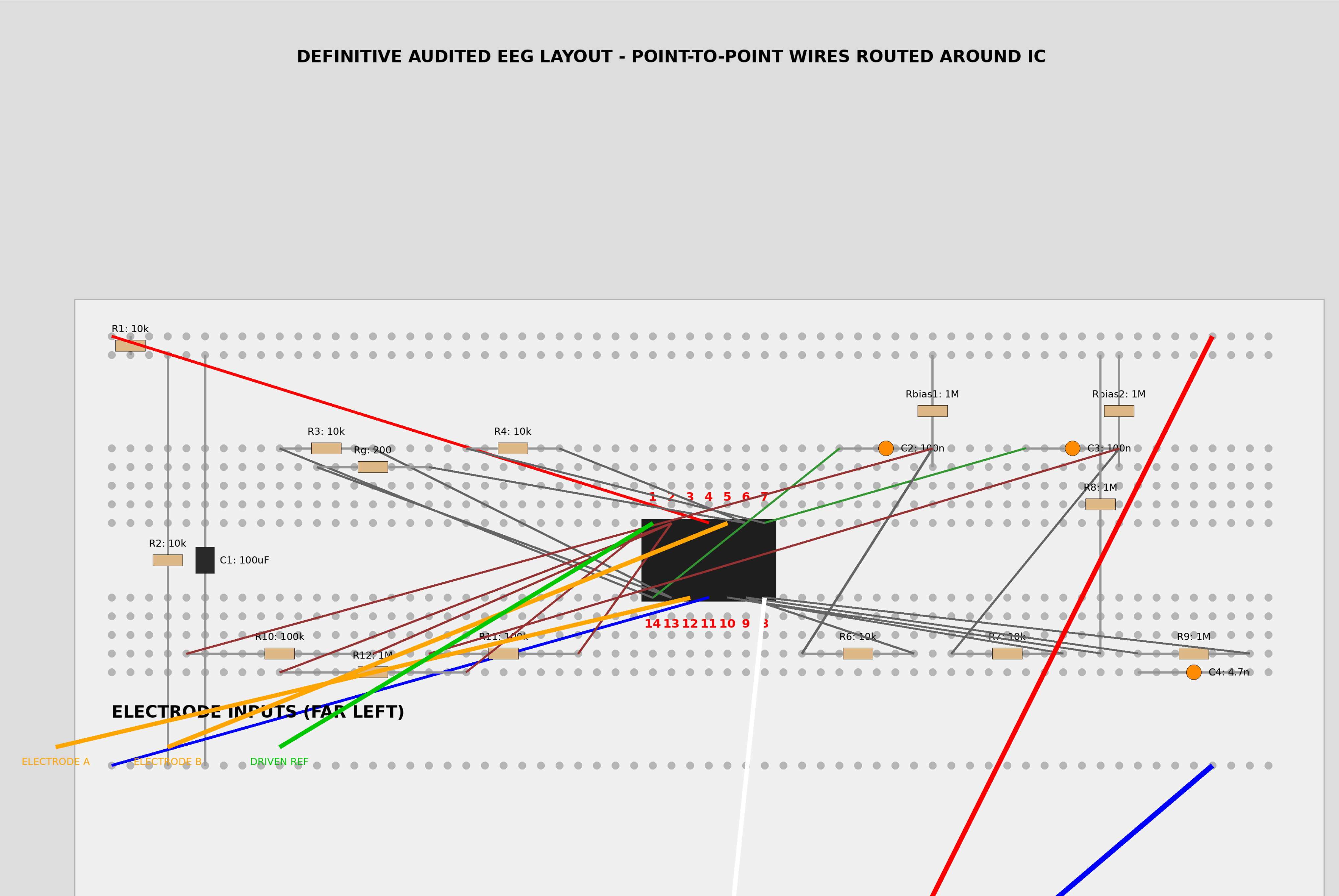

DESIGN

EEG Circuit Design: Professional DRL Redesign (10,000x Gain)

Technical Architecture

1. Precision Input buffers: High-impedance buffering for Electrode A and B.

2. AC Inter-stage Coupling: 100nF and 1M bias resistors block DC offsets.

3. Difference & Gain Stage: 10,000x total gain (100x Stage 1, 100x Stage 2).

4. Active Noise Cancellation: Driven-Right-Leg (DRL) circuit cancels hum.

5. Power: Single 9V battery with 4.5V Virtual Ground.

ASSEMBLY

Assembly & Safety Guide

- Battery: Under board.

- IC: Center.

- Electrodes: Left.

- Scope: Bottom.

TESTING

Safety & Alpha Wave Testing

Safety

- Battery power only is mandatory.

- DRL current is limited to <10uA by 1M series resistor.

testing

- Close eyes to see rhythmic Alpha oscillations (8-13Hz).

CONNECTIONS

Verified Exhaustive Connection List (Active DRL Subtractor)

1. Power

- Pin 4: V+ (+9V)
- Pin 11: 0V (Battery Negative)
- R1 (10k): V+ to VGND rail
- R2 (10k): VGND rail to 0V rail
- C1 (100uF): VGND to 0V

2. Signal Chain

- Electrode A -> Pin 12 (U1A In+)
- Pin 13 (In-) -> Pin 14 (Out) - Unity Gain Buffer
- Electrode B -> Pin 5 (U1B In+)
- Pin 6 (In-) -> Pin 7 (Out) - Unity Gain Buffer

3. Difference stage & Gain

- R3 (10k): Pin 14 to Pin 9 (U1D In-)
- R4 (10k): Pin 7 to Pin 10 (U1D In+)
- R5 (1M): Pin 10 to VGND
- Rf (1M): Pin 9 to Pin 8 (Out)
- Cf (4.7nF): Pin 9 to Pin 8

4. Active DRL

- R6 (100k): Pin 14 to Pin 2 (U1C In-)

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- R7 (100k): Pin 7 to Pin 2 (U1C In-)

- Rdrl (1M): Pin 2 to Pin 1 (Out)

- Pin 3 (In+): To VGND

- Pin 1 (DRL Out) -> Driven Reference Electrode